

CPSC 544

Advanced Software Process

Part I: Software Process Maturity

3. Software Process Assessment

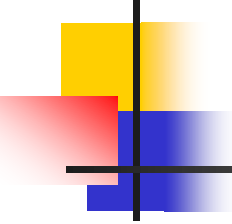
Chang-Hyun Jo

Professor, PhD

Department of Computer Science

California State University Fullerton

<http://jo.ecs.fullerton.edu>



Outlines

Part I: Software Process Maturity

1. A Software Maturity Framework
2. The Principles of Software Process Change
3. Software Process Assessment
4. The Initial Process

Part II: The Repeatable Process

5. Managing Software Organizations
6. The Project Plan
7. Software Configuration Management (Part I)
8. Software Quality Assurance

Part III: Defined Process

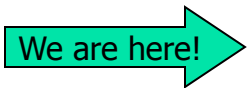
9. Software Standards
10. Software Inspections
11. Software Testing
12. Software Configuration Management (continued)
13. Defining the Software Process
14. The Software Engineering Process Group

Part IV: The Managed Process

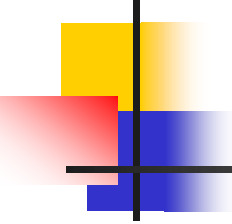
15. Data Gathering and Analysis
16. Managing Software Quality

Part V: The Optimizing Process

17. Defect Prevention
18. Automating the Software Process
19. Contracting for Software
20. Conclusion

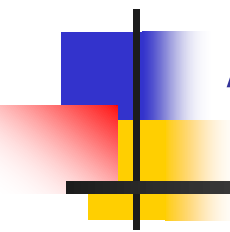


We are here!

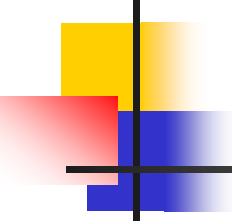


Software Process Assessment

- Assessment Overview
- Assessment Phases
- Five Assessment Principles
- The Assessment Process
- Assessment Conduct
- Implementation Considerations

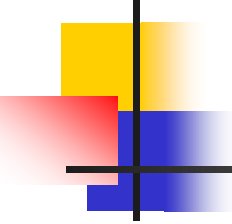


Assessment Overview



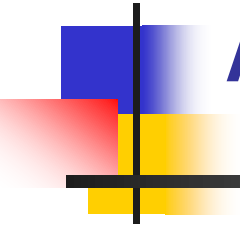
Introduction

- **Process assessment** helps SW organizations improve themselves by **identifying their critical problems** and **establishing improvement priorities**.
- The basic **assessment** objectives are:
 - To learn how the organization works
 - To identify its major problems
 - To enroll its opinion leaders in the change process

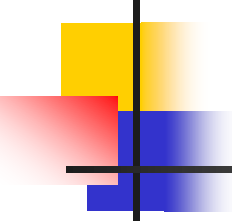


Assessment Overview

- A SW process assessment is not an audit but a review of a SW organization to advise its management and professionals on how they can improve their operation.
- It is conducted by a team of SW professionals who typically have assessment experience or training.

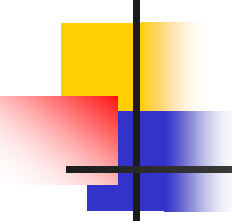


Assessment Phases



Assessment Phases

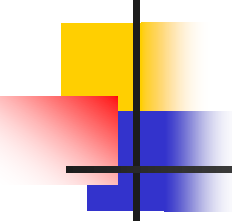
- Assessment typically conducted in three phases:
 - Preparation
 - Assessment
 - Recommendations



Assessment Phases

- Preparation

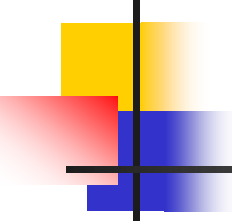
- Senior management becomes committed to the process, agree to participate personally, and commits to take action on the resulting recommendations or explain its reasons for not doing so.
- Phase one concludes with a brief one- or two-day training program for the assessment team.



Assessment Phases

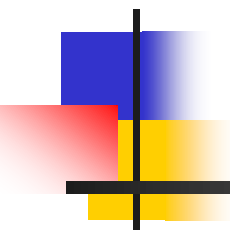
- Assessment

- Phase two is the **on-site assessment** period.
- Assessment activity typically takes several days, although it can take two or more weeks, depending on the size of the organization and the assessment technique used.
- Phase two **concludes with a preliminary report of the findings** to local management.

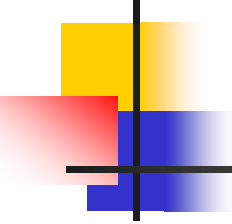


Assessment Phases

- Recommendations
 - In phase three the findings and action recommendations are presented to the local managers.
 - A local action team is then assembled to plan and implement the recommendations.
 - The assessing organization may provide assistance during this period and may participate in a subsequent follow-up assessment.

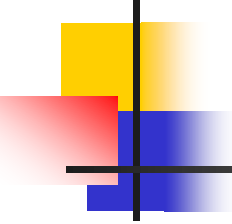


Five Assessment Principles



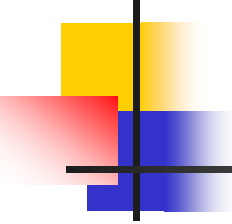
Five Assessment Principles

- As in many activities, the basic requirements for a good assessment are a competent team, sound leadership, and a cooperative organization.
- Because the SW process is human-intensive, however, some special considerations should be kept in mind.



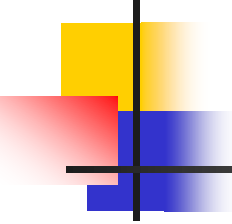
Five Assessment Principles

- The need for a **process model** as a basis for the assessment
- The requirement for **confidentiality**
- **Senior management** involvement
- **An attitude of respect** for the views of the people in the organization being assessed
- An **action** orientation



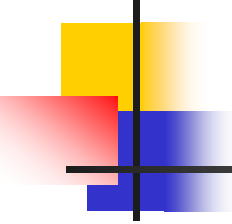
Start with a Process Model

- An **assessment** implies a **standard**.
- The organization's process is **reviewed in comparison with some vision** of how such processes should be performed.



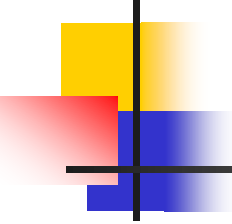
Observe Strict Confidentiality

- The assessment's purpose must be to support the organization's improvement program and **not** to report its problems to higher management.
- Even when initiated with this intent, it is extraordinarily **difficult to maintain confidentiality**, particularly when a chief executive demands to see the results.



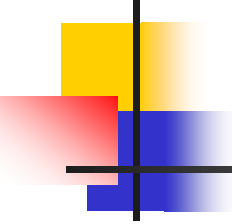
Observe Strict Confidentiality

- If assessment team provides such data, however, people will learn that they can **not** speak in confidence.
- As this becomes widely known, the assessment group will find it **increasingly difficult to conduct assessments that uncover the real issues.**
- **Confidentiality** is required at all organizational levels.



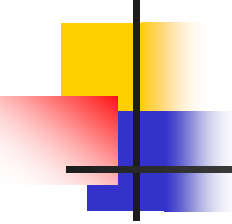
Involve Senior Management

- The senior manager sets the **organization's priorities**.
- Site manager
 - Senior manager of total organization
 - The site manager **must be personally involved in the assessment** and its follow-up action plans.
 - If not, the work will not be given sufficiently priority.



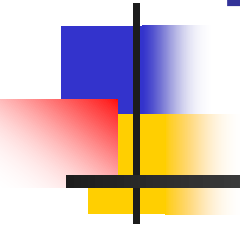
Keep an Open Mind and a Level Head

- Even when the assessment team is appropriately supportive, some local people will be **resentful** and **not cooperate**.
- If the team members' actions clearly **demonstrate their desire for active collaboration** with the on-site professionals, however, this will be recognized and people generally will respond positively.

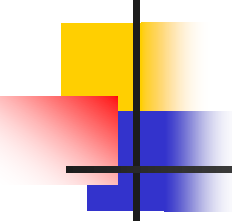


Focus on Action

- The **assessment** must be directly toward **improvement**.
- An **action orientation** keeps the questions focused on current problems and the need to solve them.
- If the assessment turns into a **general exploration**, it will **not focus on the priority issues or produce recommendations** that will be implemented.
- Management must **either focus on taking action or not do an assessment**.

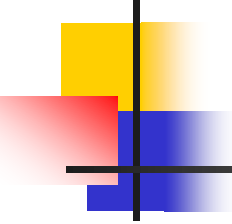


The Assessment Process



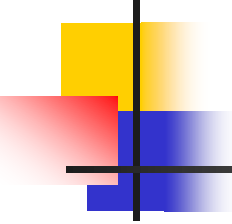
The Assessment Process

- The first step in any assessment is to identify **the organization** to be assessed and **the team** to do it.
- This typically requires the **site manager's commitment** to doing the assessment and willingness to assign sufficient skilled resources to get it done.



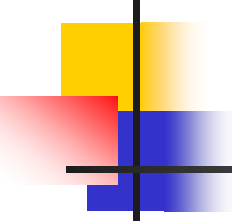
Forming an Assessment Team

- The assessment team leader is selected first.
- The assessment team members should all be experienced SW developers, and one or more should have experience in each phase of the SW process.
- Four to six professionals typically form an adequate team, although more can be used if desired.



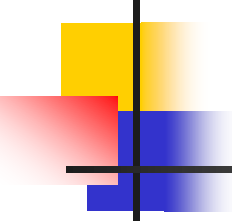
Guidelines for Selecting Assessment Team Members

- Each assessment team member should:
 - Have at least eight to ten years professional SW experience
 - Be well respected in the organization
 - Be able to deal with people in an informal and non-threatening manner
 - Be a team player
 - Have attended assessment training with this team
- No assessment team member should:
 - Be currently serving in an audit or review capacity for any of the projects being assessed
 - Be a line manager over any of the projects being assessed or people being interviewed
 - Be working directly on any of the projects being assessed or working on their direct support



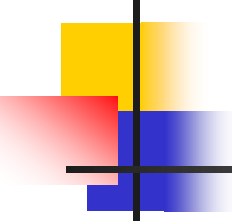
Self-Assessment Considerations

- While it is possible for organizations to **assess themselves**, they should be aware of several **potential problems**.
 - Few organizations can afford a staff of assessment experts.



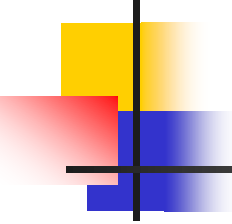
Assessment Ground Rules

- It is desirable to have a written set of assessment ground rules for the organization being assessed and for the assessment team members.
- For an external assessment, the site manager and the assessment team leader should sign a written agreement covering these ground rules.
- A copy of the standard agreement used by the SEI is shown in Appendix B.



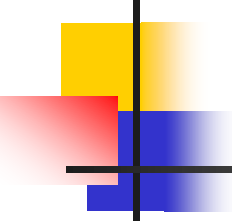
Assessment Team Training

- As the **assessment team** is formed, the **members must agree to participate fully** during the training period, on-site review, and wrap-up meetings.



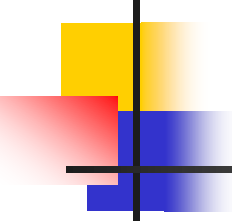
The On-Site Period

- The **assessment** starts with a **presentation** to the site manager and staff.
- The **assessment ground rules** are discussed, as well as the **assessment principles** and the **overall schedule**.
- An **overview meeting** is then held with all the site participants, including the project managers and the professionals to be interviewed.



Typical SEI-Assisted Assessment Schedule [Table 3.2]

- Day One
 - Assessment Overview, Briefing, Questionnaire, Project Discussions
- Day Two
 - Functional Area Interviews, Preliminary Findings
- Day Three
 - Project Discussions, Finding Formulation
- Day Four
 - Findings Dry Run, Findings Review, Findings Presentation, Senior Management Meeting, Assessment Postmortem



Assessment Conduct

- The **assessment** should focus on **what the projects actually do, how they do it, the problems encountered, and the results obtained.**
- A **selected set of questions** should be prepared in advance of the actual assessment period.
- **Meetings** are next held with small group of selected professionals who have **expertise in various facets of the SW process.**
 - **Table 3.3** Guidelines for Selecting Functional Area Representatives

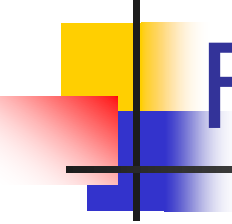
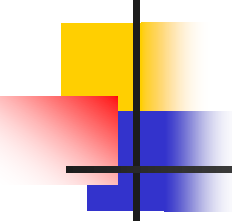


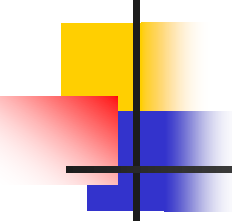
Table 3.3 Guidelines for Selecting Functional Area Representatives

- Six to eight representatives are identified for each of several meetings of 90 to 120 minutes each.
- Potential areas of specialty for each meetings are:
 - Quality Assurance and release
 - SW integration and test
 - Coding and unit test
 - Requirements and design
- Each of these representative should be:
 - A recognized opinion leader in the selected area
 - Actually working on projects rather than on a staff
 - A technical professional and not a manager
 - Aware of the assessment process and the confidentiality arrangements



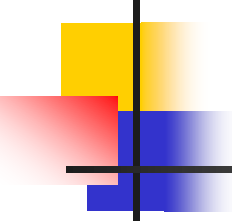
Probing Questions

- In conducting assessments it is hard to obtain really accurate information.
 - Questions are often misunderstood.
 - The respondents may have a different understandings of some common terms.
 - The respondents may not be broadly aware of the work in their own organization.
 - Occasionally people are unwilling to risk the truth.



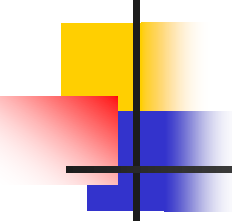
Assessment Conclusions

- At the **assessment conclusion**, the team prepares **a report on its initial findings**.
 - **Table 3.4 Guidelines for Finding Formulation**
- The report should be **a composite summary of site status**, together with more detailed findings in key areas.
- Prior to **reviewing** this material with **the site manager**, the team should review it with **the project managers**.
 - This should identify any overlooked problems or any misstated or overemphasized topics.



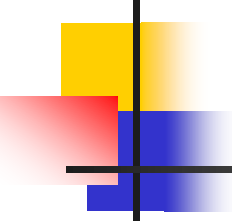
Assessment Conclusions

- The last site actions is to review the composite findings with the site manager and staff.
- Any questions are addressed, and the schedule for follow-up work is reviewed.
- If possible, all the site personnel who participated in the assessment should attend this meeting.
- Table 3.5 A Typical Outline of the Findings Presentation



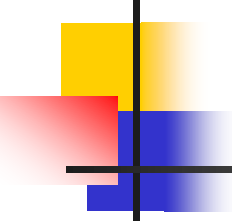
Guidelines for Finding Formulation [Table 3.4]

- The findings should be limited to the top 10 to 12 items that are:
 - Major issues for most of the projects reviewed
 - Key issues for most of the projects reviewed
 - Key issues for advancing to the next maturity level
 - Supported by evidence from the assessment
 - Addressable by an action recommendation
 - Specific – avoid sweeping generalizations



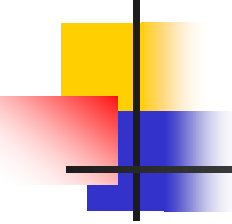
A Typical Outline of the Findings Presentation [Table 3.5]

- Presentation Overview:
 - Agenda
 - Scope – projects assessed
 - Assessment conduct – schedule
 - Summary composite status – the bad news
 - Strengths noted – it is not all bad
 - Findings summary
 - Findings – each finding
 - Next steps
- Findings Format:
 - Finding – what the team observed
 - Consequences – the implications of the finding for the organization
 - Examples – specific instances of the finding (without identifying people or projects)



Assessment Report

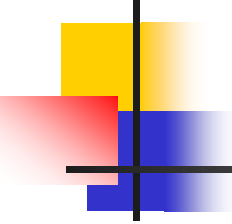
- The final assessment team action is the presentation of a written final report and recommendations to the site manager and staff.
 - The recommendations should highlight the three or four items of highest priority.
 - Since no organization can handle more than a few priority tasks at a time, the total number of items requiring attention should be limited to around ten.
 - These should be clearly explained with the assessment team's views on implementation priority.



A Typical Final Report Outline

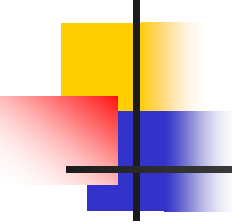
[Table 3.6]

- Summary and Conclusions
 - An executive summary of the findings and recommendations
- The Assessment
 - The assessment background and a description of the assessment process
- Site Status
 - A summary description of the site status
- Key Findings
 - A brief description of each finding
- Recommendations
 - A description of each of the recommendations in priority order
- Appendices
 - The assessment agreement
 - Cross-reference of findings and recommendations



Action Plans

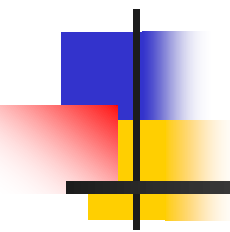
- The **action plans** are next **prepared by the local site organization**, generally under the guidance of the **team member** named for this purpose.
- If properly chosen, this **member** is now fully knowledgeable on the issues and is able to start quickly.

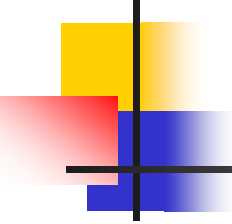


Reassessments

- Organizations should generally conduct **follow-up assessment one to two years after the initial action plans** have been developed and approved.
- This is important for several reasons:
 - To assess the progress that has been made
 - To provide a visible future milestone for completion of the actions from the prior assessment
 - To establish new priorities for continued improvement

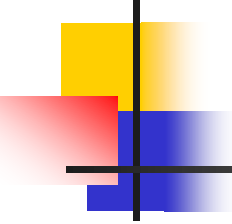
Implementation Considerations





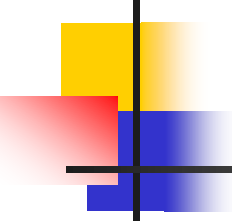
Implementation Considerations

- The greatest **assessment risk** is that **no significant improvement actions** will be taken.
- Without proper management focus a few superficial efforts may be made, but soon everything will revert to business as usual.



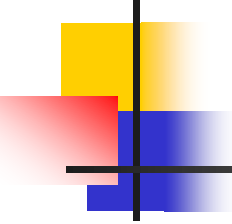
Implementation Considerations

- A catalyst is needed to maintain the improvement priority, such as **goals** and **management reviews**.
 - **Long-term goals** are first established, and then **sub-goals** are defined for intervening **two- or three-month periods**.
 - A **senior management quarterly review** then maintains high-level checkpoint visibility, crystallizes the plans, and creates the periodic crises required to get things accomplished.



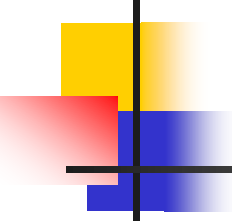
Risks

- Some of **the other key risks** and potential actions to alleviate them are **schedule conflicts, inadequate support, and lack of follow-through.**



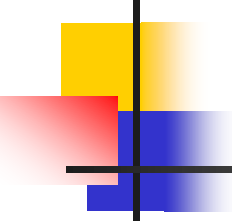
Risks: Schedule Conflicts

- Despite the best intentions, **crises that conflict with assessment plans** often arise.
- The most damaging of these require the site manager to miss either the opening or the closing meeting or both.
 - Such unfortunate schedule conflicts have **happened in about one-third of the assessments.**
 - A solution - substitute executive who can speak for the site manager and then arrange for a later private meeting to cover the issues



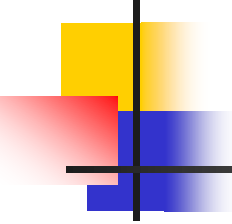
Risks: Inadequate Support

- The reason of **inadequate support** is that the assessment commitment is **made at too low a management level**.
- Often only a very senior executive can **take a sufficiently long-term view to avoid becoming defensive**.
- Also, even fairly high-level managers are often **only responsible for portions**, so they **cannot** provide adequate **organization-wide** priority.



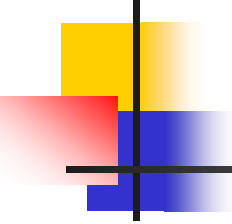
Risks: Lack of Follow-through

- Frequently management change or other high-priority issues reduce the focus on action plan implementation.
- The most important determinants of success is the presence of an aggressive manager to lead the change efforts, a capable process improvement staff, and a clearly stated improvement goal.



Staffing

- Staffing is generally the most serious implementation problem.
- In addition to staffing the assessment itself, the other staffing needs will generally include:
 - A small, full-time staff to focus and to guide the improvement efforts
 - Part-time project participation in the action plan working group
 - Project review and implementation of the resulting actions



References

- Humphrey, Watts S., *Managing the Software Process*, The SEI Series in Software Engineering, Addison-Wesley, 1989. (29th Printing, May 2003) (ISBN 0-201-18095-2)